

CBSE Class 10 Mathematics — Triangles

Time Allowed: 3 Hours

Maximum Marks: 80

General Instructions:

1. This question paper contains questions. All questions are compulsory.
1. Questions are grouped into sections A through E.
2. Use of calculators is NOT permitted.

Section A: Multiple Choice & Assertion-Reason (1 mark each)

1. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 2$ cm, $DB = 3$ cm, and $AE = 3$ cm. Find the length of EC .
(A) (A) 3.5 cm
(B) (B) 4 cm
(C) (C) 4.5 cm
(D) (D) 5 cm
2. If $\triangle ABC \sim \triangle PQR$ and $\frac{AB}{PQ} = \frac{1}{3}$, then the ratio of their areas $\frac{\text{Area}(\triangle ABC)}{\text{Area}(\triangle PQR)}$ is:
(A) (A) 1:3
(B) (B) 1:9
(C) (C) 3:1
(D) (D) 9:1
3. In $\triangle ABC$, if $AB = 6\sqrt{3}$ cm, $AC = 12$ cm, and $BC = 6$ cm, then the measure of $\angle B$ is:
(A) (A) 45°
(B) (B) 60°
(C) (C) 30°
(D) (D) 90°
4. Two triangles are similar if their corresponding sides are in the same ratio. This criterion is known as:
(A) (A) SSS
(B) (B) SAS
(C) (C) AA
(D) (D) RHS

5. The altitude of an equilateral triangle with side $2a$ is:
- (A) (A) a
 - (B) (B) $a\sqrt{2}$
 - (C) (C) $a\sqrt{3}$
 - (D) (D) $2a$
6. Assertion (A): If a line divides any two sides of a triangle in the same ratio, then the line is parallel to the third side. Reason (R): This is the Converse of Basic Proportionality Theorem.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
7. Assertion (A): Two triangles are similar if their corresponding angles are equal. Reason (R): If two triangles are equiangular, they are similar by AAA similarity criterion.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
8. Assertion (A): In $\triangle ABC$, if $DE \parallel BC$ such that $AD/DB = 1/2$ and $AE = 2$ cm, then $EC = 4$ cm. Reason (R): By Basic Proportionality Theorem, if $DE \parallel BC$, then $AD/DB = AE/EC$.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 - (C) (C) Assertion (A) is true but Reason (R) is false
 - (D) (D) Assertion (A) is false but Reason (R) is true
9. Assertion (A): All congruent triangles are similar. Reason (R): All similar triangles are congruent.
- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 - (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)

- (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true

10. Assertion (A): The ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides. Reason (R): This is the Area of Similar Triangles Theorem.

- (A) (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A)
 (B) (B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A)
 (C) (C) Assertion (A) is true but Reason (R) is false
 (D) (D) Assertion (A) is false but Reason (R) is true

Section C: Short Answer I (3 marks each)

1. In $\triangle ABC$, $DE \parallel BC$ such that $AD = 2.4$ cm, $AE = 3.2$ cm, and $EC = 4.8$ cm. Find the length of DB .
2. Two triangles $\triangle ABC$ and $\triangle DEF$ are similar. If $Area(\triangle ABC) = 64$ cm² and $Area(\triangle DEF) = 121$ cm², and $EF = 15.4$ cm, find the length of side BC .
3. A vertical pole of length 6 m casts a shadow 4 m long on the ground and at the same time a tower casts a shadow 28 m long. Find the height of the tower.
4. In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $\angle PST = \angle PRQ$. If $PS = 2$ cm, $SQ = 3$ cm, and $PT = 2.5$ cm, find TR .
5. Diagonals of a trapezium $ABCD$ with $AB \parallel DC$ intersect each other at the point O . If $AB = 2CD$, find the ratio of the areas of triangles $\triangle AOB$ and $\triangle COD$.

Section D: Long Answer (4-5 marks each)

1. In $\triangle ABC$, $DE \parallel BC$ such that $AD = (4x - 3)$ cm, $AE = (8x - 7)$ cm, $BD = (3x - 1)$ cm, and $EC = (5x - 3)$ cm. Find the value of x .
2. Two poles of heights 6 m and 11 m stand on a plane ground. If the distance between their feet is 12 m, find the distance between their tops.
3. In $\triangle PQR$, S and T are points on sides PQ and PR respectively such that $\angle PST = \angle PRQ$. Show that $\triangle PQR \sim \triangle PTS$.
4. The areas of two similar triangles are 121 cm² and 64 cm² respectively. If the median of the first triangle is 12.1 cm, find the corresponding median of the other triangle.
5. In $\triangle ABC$, $AD \perp BC$ such that $AD^2 = BD \cdot CD$. Prove that $\angle BAC = 90^\circ$.
6. State and prove the Basic Proportionality Theorem (Thales Theorem).

7. Prove that the ratio of the areas of two similar triangles is equal to the square of the ratio of their corresponding sides.
8. In $\triangle ABC$, $AD \perp BC$ such that $AD^2 = BD \cdot CD$. Prove that $\triangle ABC$ is a right-angled triangle at A .
9. A vertical stick 2 m long casts a shadow 1 m long on the ground. At the same time, a tower casts a shadow 5 m long. Find the height of the tower using the properties of similar triangles.